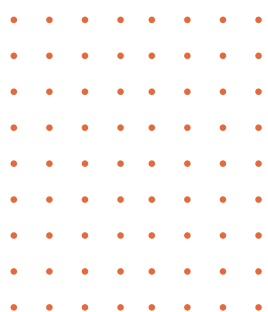




VIRTUAL INTERNSHIP

Variance Analysis

Task 1



01

Order Quantity

Built using DAX Measures

| Power Bi | Dax |



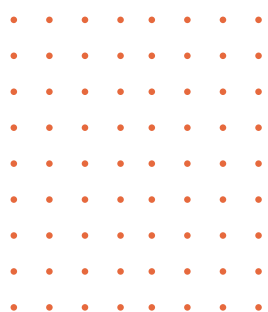
DAX MEASURES ORDER QUANTITY

BM_ORDER_QTY = SUM(BENCHMARKS[TOTAL_ORDER_QUANTITY])

RECORDED_ORDER_QTY = SUM(FACT_ORDERS[ORDER_QTY])

DIFF_ORDER_QTY = ABS([RECORDED_ORDER_QTY] - [BM_ORDER_QTY])

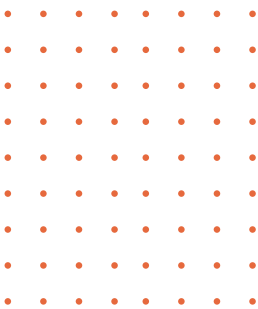
ORDER_%_BM = $\frac{\text{ABS}([\text{RECORDED_ORDER_QTY}] - [\text{BM_ORDER_QTY}])}{[\text{BM_ORDER_QTY}] * 100}$





Order Quantity Variance Analysis Report

mmm_yy	customer_name	customer_id	First city	BM_Order_Qty	Recorded_Order_Qty	Diff_Order_Qty	Order_%_BM
Mar_22	Acclaimed Stores	789520	Surat	64298	63367	931	1.45
Mar_22	Acclaimed Stores	789521	Ahmedabad	62069	63216	1147	1.85
Mar_22	Acclaimed Stores	789522	Vadodara	62200	61813	387	0.62
Mar_22	Atlas Stores	789220	Surat	62905	61784	1121	1.78
Mar_22	Atlas Stores	789221	Ahmedabad	62281	62116	165	0.26
Mar_22	Chiptec Stores	789320	Surat	66900	67092	192	0.29
Mar_22	Chiptec Stores	789321	Ahmedabad	68464	67626	838	1.22
Mar_22	Coolblue	789121	Ahmedabad	64186	64786	600	0.93
Mar_22	Coolblue	789122	Vadodara	70465	69763	702	1.00
Mar_22	Elite Mart	789902	Ahmedabad	72049	72967	918	1.27
Mar_22	Elite Mart	789903	Vadodara	63885	63138	747	1.17
Mar_22	Expert Mart	789621	Ahmedabad	67690	67583	107	0.16
Mar_22	Expert Mart	789622	Vadodara	71241	72320	1079	1.51
Mar_22	Expression Stores	789301	Surat	67439	66567	872	1.29
Mar_22	Expression Stores	789303	Vadodara	64694	64591	103	0.16
Mar_22	Info Stores	789601	Surat	66860	67092	232	0.35
Total			Ahmedabad	13426936	13404261	22675	0.17



02

Delivery Quantity

Built using DAX Measures

| Power Bi | Dax |



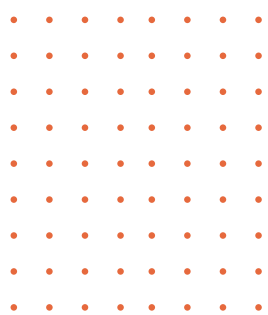
DAX MEASURES DELIVERY QUANTITY

BM_DELIVERY_QTY = SUM(BENCHMARKS[TOTAL_DELIVERY_QUANTITY])

RECORDED_DELIVERY_QTY = SUM(FACT_ORDERS[DELIVERY_QTY])

DIFF_DELIVERY_QTY = ABS([RECORDED_DELIVERY_QTY] - [BM_DELIVERY_QTY])

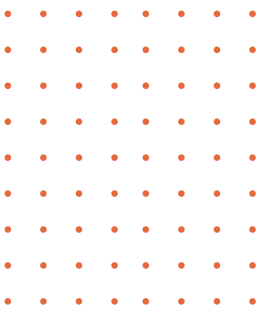
DELIVERY_%_BM = $\frac{\text{ABS}([RECORDED_DELIVERY_QTY] - [BM_DELIVERY_QTY])}{[BM_DELIVERY_QTY]} * 100$





Delivery Quantity Variance Analysis Report

mmm_yy	customer_name	customer_id	city	BM_Delivery_Qty	Recorded_Delivery_Qty	Diff_Delivery_Qty	Delivery_%_BM
Mar_22	Acclaimed Stores	789520	Surat	59755	60098	343	0.57
Mar_22	Acclaimed Stores	789521	Ahmedabad	60432	60461	29	0.05
Mar_22	Acclaimed Stores	789522	Vadodara	60480	60403	77	0.13
Mar_22	Atlas Stores	789220	Surat	61465	61253	212	0.34
Mar_22	Atlas Stores	789221	Ahmedabad	60787	59527	1260	2.07
Mar_22	Chiptec Stores	789320	Surat	65042	65625	583	0.90
Mar_22	Chiptec Stores	789321	Ahmedabad	66706	66590	116	0.17
Mar_22	Coolblue	789121	Ahmedabad	62549	62349	200	0.32
Mar_22	Coolblue	789122	Vadodara	65821	66376	555	0.84
Mar_22	Elite Mart	789902	Ahmedabad	70389	70998	609	0.87
Mar_22	Elite Mart	789903	Vadodara	59324	59160	164	0.28
Mar_22	Expert Mart	789621	Ahmedabad	66132	66103	29	0.04
Mar_22	Expert Mart	789622	Vadodara	69291	69465	174	0.25
Mar_22	Expression Stores	789301	Surat	65402	65857	455	0.70
Mar_22	Expression Stores	789303	Vadodara	63551	63901	350	0.55
Mar_22	Info Stores	789601	Surat	61949	60551	1398	2.26
Total				12969157	12950496	18661	0.14

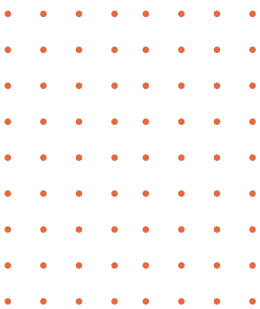




1. WHICH CUSTOMER_ID HAS THE LARGEST ABSOLUTE DIFFERENCE BETWEEN THEIR RECORDED ORDER QUANTITY AND THE BENCHMARK ORDER QUANTITY?

MAY 22 - ELITE MART - 789903 - VADODARA

Order Quantity Variance Analysis Report							
mmm_yy	customer_name	customer_id	First city	BM_Order_Qty	Recorded_Order_Qty	Diff_Order_Qty	Order_%_BM
May_22	Elite Mart	789903	Vadodara	63918	68699	4781	7.48
May_22	Sorefoz Mart	789702	Ahmedabad	69569	65568	4001	5.75
May_22	Chiptec Stores	789321	Ahmedabad	61590	57592	3998	6.49





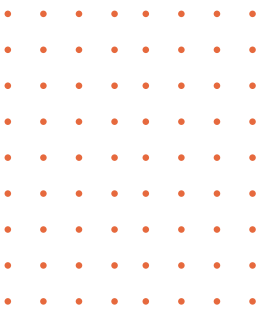
2.HOW MANY CUSTOMERS IN THE DELIVERY CATEGORY HAVE A DIFFERENCE OF GREATER THAN 3% BETWEEN THEIR RECORDED DELIVERY QUANTITY AND THE BENCHMARK DELIVERY QUANTITY, EXPRESSED AS A PERCENTAGE OF THE BENCHMARK?

ANSWER : 05

Delivery Quantity Variance Analysis Report

mmm_yy	customer_name	customer_id	city	BM_Delivery_Qty	Recorded_Delivery_Qty	Diff_Delivery_Qty	Delivery_%_BM
Jul_22	Lotus Mart	789420	Surat	62968	58046	4922	7.82
Jun_22	Lotus Mart	789420	Surat	58213	55235	2978	5.12
Jul_22	Elite Mart	789903	Vadodara	60714	58378	2336	3.85
Jul_22	Vijay Stores	789102	Ahmedabad	69855	72041	2186	3.13
Jul_22	Rel Fresh	789203	Vadodara	69070	66932	2138	3.10
May_22	Lotus Mart	789420	Surat	68359	70322	1963	2.87
Jul_22	Acclaimed Stores	789522	Vadodara	63019	64793	1774	2.82
Aug_22	Viveks Stores	789501	Surat	58321	56669	1652	2.83
Jul_22	Propel Mart	789401	Surat	59995	61609	1614	2.69
Aug_22	Viveks Stores	789503	Vadodara	59832	58240	1592	2.66
Jun_22	Propel Mart	789403	Vadodara	61997	60445	1552	2.50
Jul_22	Rel Fresh	789202	Ahmedabad	63941	65491	1550	2.42

Counted at customer month level. Lotus Mart (789420) appears twice in Jun_22 and Jul_22 as it independently crossed the 3% threshold in both months.

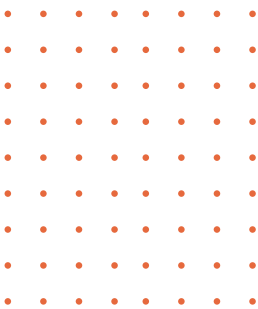




3.WHAT IS THE QUANTITY OF ORDERS RECORDED FOR “VIVEKS STORES” IN THE CITY OF 'VADODARA' DURING THE MONTH OF MARCH 2022?

ANSWER : 73011

Order Quantity Variance Analysis Report								
Ahmedabad	mmm_yy	customer_name	customer_id	First city	BM_Order_Qty	Recorded_Order_Qty	Diff_Order_Qty	Order_%_BM
Surat	Mar_22	Viveks Stores	789503	Vadodara	73308	73011	297	0.41
Vadodara	Mar_22	Expert Mart	789622	Vadodara	71241	72320	1079	1.51
Apr_22	Mar_22	Coolblue	789122	Vadodara	70465	69763	702	1.00
Jun_22	Mar_22	Lotus Mart	789422	Vadodara	64849	66331	1482	2.29
Aug_22	Mar_22	Expression Stores	789303	Vadodara	64694	64591	103	0.16
Mar_22	Mar_22	Info Stores	789603	Vadodara	64551	63493	1058	1.64
May_22	Mar_22	Elite Mart	789903	Vadodara	63885	63138	747	1.17
	Mar_22	Vijay Stores	789103	Vadodara	63098	63396	298	0.47
	Mar_22	Acclaimed Stores	789522	Vadodara	62200	61813	387	0.62
	Mar_22	Sorefoz Mart	789703	Vadodara	61951	61102	849	1.37
	Mar_22	Propel Mart	789403	Vadodara	61465	62734	1269	2.06





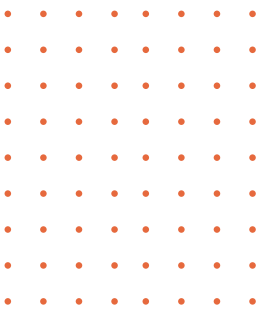
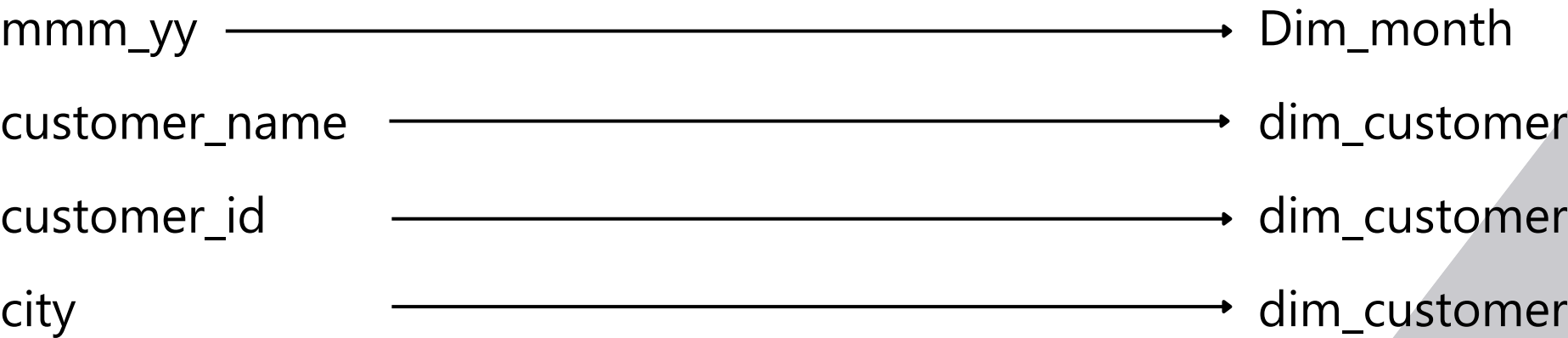
I CREATED A NEW TABLE CALLED DIM_MONTH

MODELING TAB > NEW TABLE THEN THE DAX

Dim_Month =

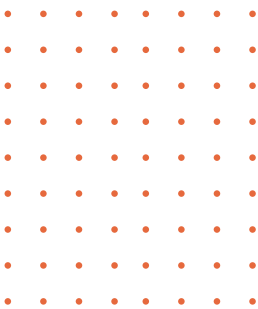
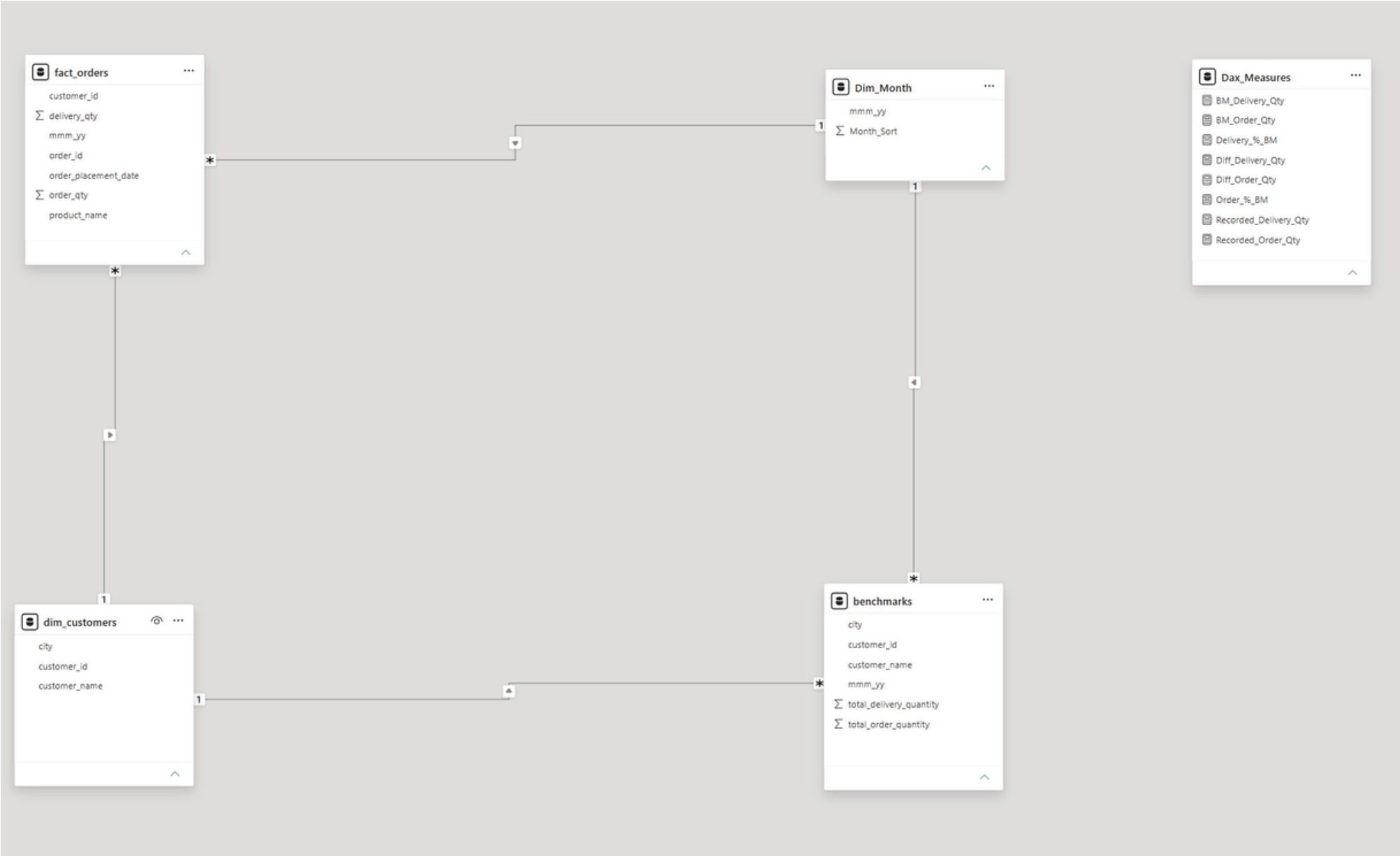
```
ADDCOLUMNS(
  DISTINCT(UNION(VALUE(FACT_ORDERS[MMM_YY]),
    VALUE(BENCHMARKS[MMM_YY]))),
  "MONTH_SORT",
  SWITCH(
    [MMM_YY],
    "MAR_22", 1,
    "APR_22", 2,
    "MAY_22", 3,
    "JUN_22", 4,
    "JUL_22", 5,
    "AUG_22", 6 )
)
```

COMMON COLUMNS PULLED FROM THE TABLES





ER DIAGRAM



Linkedin Profile :

<https://www.linkedin.com/in/vishnu-ram-sachin-d-/>